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INTERCONNECT STRUCTURE

Abstract

A semiconductor structure includes a transistor, a conductive via extending through a dielectric layer and electrically connected to the transistor, a metal cap disposed on a first surface of the conductive via and a first portion of a second surface of the dielectric layer, and a metal line disposed around the metal cap. The metal line includes a barrier layer and a metal fill layer embedded in the barrier layer. The metal cap is embedded in the metal line. The barrier layer extends along a second portion of the second surface of the dielectric layer and is spaced apart from the conductive via.

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Background/Summary

PRIORITY [0001] The present application is a continuation application of U.S. patent application Ser. No. 17/884,263, filed on Aug. 9, 2022, which is a divisional application of U.S. patent application Ser. No. 17/104,760, filed on Nov. 25, 2020, now issued as U.S. Pat. No. 11,670,581, each of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

[0003] For example, to prevent deterioration of vias or metal lines due to contact with adjacent dielectric layer, the vias or metal lines may be lined with a barrier layer. While existing interconnect structures are generally adequate for their intended purposes, they are not satisfactory in all aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 illustrates a flowchart of a method of forming an exemplary interconnect structure or a portion thereof, in accordance with one or more aspects of the disclosure.

[0006] FIG. 2 illustrates a three-dimensional (3D) perspective view of a workpiece undergoing various stages of operations in the method of FIG. 1 in accordance with one or more aspects of the disclosure.

[0007] FIGS. 3-14 illustrate fragmentary cross-sectional views of a portion of the workpiece undergoing various stages of operations in the method of FIG. 1 in accordance with one or more aspects of the present disclosure.

DETAILED DESCRIPTION

[0008] It is understood that the following disclosure provides many different embodiments, or examples, for implementing different features of the present disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of

course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments, in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the sake of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Moreover, various features may be arbitrarily drawn in different scales for the sake of simplicity and clarity.

[0009] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as being “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0010] Still further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range considering variations that inherently arise during manufacturing as understood by one of ordinary skill in the art. For example, the number or range of numbers encompasses a reasonable range including the number described, such as within $\pm 10\%$ of the number described, based on known manufacturing tolerances associated with manufacturing a feature having a characteristic associated with the number. For example, a material layer having a thickness of “about 5 nm” can encompass a dimension range from 4.25 nm to 5.75 nm where manufacturing tolerances associated with depositing the material layer are known to be $\pm 15\%$ by one of ordinary skill in the art. Still further, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0011] IC manufacturing process flow is typically divided into three categories: front-end-of-line (FEOL), middle-end-of-line (MEOL), and back-end-of-line (BEOL). FEOL generally encompasses processes related to fabricating IC devices, such as transistors. For example, FEOL processes can include forming active regions (such as fins), gate structures, and source and drain features (generally referred to as source/drain features). MEOL generally encompasses processes related to fabricating contacts to conductive features (or conductive regions) of the IC devices, such as contacts to the gate structures and/or the source/drain features. BEOL generally encompasses processes related to fabricating a multilayer interconnect (MLI) feature that interconnects IC features fabricated by FEOL and MEOL (referred to herein as FEOL and MEOL features or structures, respectively), thereby enabling operation of the IC devices.

[0012] The present disclosure generally relates to reducing or substantially eliminating electromigration and, more particularly, to selective deposition of a metal cap layer above a contact via to mitigate metal migration and reduce contact resistance.

[0013] The IC fabrication industry has gone over different conductive materials to improve performance of interconnect structures. For example, aluminum was once a popular conductive material for construction of interconnect structure because of its ready integration with IC fabrication processes and its improved conductivity as compared to doped polysilicon. Due to its susceptibility to electromigration and inferior conductivity compared to copper, aluminum fell out of favor and was gradually replaced by copper, despite the various difficulties in integrating copper

into IC fabrication processes. However, while copper is less susceptible to electromigration, it is not immune to it. Electromigration refers to material migration due to presence of current flows. The moving electrons in the current flow are scattered by metal ions in the crystal lattice and their momentum may be transferred to the metal ions. This momentum transfer from the electrons to the metal ions creates a momentum transfer force. When this momentum transfer force is greater than an activation energy of the conductor, a diffusion process of the metal ions along the direction of the moving electrons may be resulted, hence electromigration. Migration of metal atoms along different diffusion paths may result in depletion or accumulation of metal. Depletion is manifested as voids or vacancies and accumulation is manifested as whiskers or hillocks. The growth of voids may cause an onset of a positive feedback loop that accelerates formation of defects. In the positive feedback loop, voids may cause current crowding and increase local current density near the voids, the local current density increase may lead to Joule heating which may cause lowering of activation energies, and the lowered activation energies in turn speed up the material diffusion.

[0014] Conductive features (e.g., contact vias and/or metal lines) may be fabricated using, for example, Damascene processes or Dual-Damascene processes. In an exemplary Damascene process, an opening (a via opening or a trench) is formed in a dielectric layer using lithography and etching processes. Copper is then deposited in the opening and a planarization process is performed to remove excess copper, leaving a copper conductive feature in the opening. Allowing copper to come in contact with silicon or silicon oxide has negative implications. For example, copper may diffuse into silicon to degrade the semiconductor properties of silicon and lead to formation of less conductive copper silicide. Oxygen in silicon oxide and copper in the copper conductive layer may inter-diffuse and degrade the conductivity of the copper conductive layer. To prevent copper's contact with silicon and silicon oxide, barrier layers or liners may be implemented to encapsulate the copper conductive feature. As described above, as the geometry size and thus the interconnect structures are getting smaller, the barrier layer is getting thinner. However, the thinned barrier layer would result in metal migration between the metal line and the contact via, for example, thereby impacting device performance and lifetime. In the positive feedback loop, voids and hillocks may grow faster, and the positive feedback loop will accelerate the formation of voids and hillocks, resulting in via bond defects (VBDs), time-dependent dielectric breakdown (TDDB) and resistive-conductive (RC) delay. Here, VBDs refer to defects at the interface between a contact via and a metal line. TDDB refers to degradation of dielectrics due to copper diffusion. RC delay refers to be increase of resistance and/or conductance due to formation of voids or hillocks.

[0015] As integrated circuit (IC) technologies progress towards smaller technology nodes, multi-gate devices have been introduced to improve gate control by increasing gate-channel coupling, reducing off-state current, and reducing short-channel effects (SCEs). A multi-gate device generally refers to a device having a gate structure, or portion thereof, disposed over more than one side of a channel region. Fin-like field effect transistors (FinFETs) and multi-bridge-channel (MBC) transistors are examples of multi-gate devices that have become popular and promising candidates for high performance and low leakage applications. A FinFET has an elevated channel wrapped by a gate on more than one side (for example, the gate wraps a top and sidewalls of a “fin” of semiconductor material extending from a substrate). An MBC transistor has a gate structure that can extend, partially or fully, around a channel region to provide access to the channel region on two or more sides. Because its gate structure surrounds the channel regions, an MBC transistor may also be referred to as a surrounding gate transistor (SGT) or a gate-all-around (GAA) transistor. The channel region of an MBC transistor may be formed from nanowires, nanosheets, or other nanostructures and for that reasons, an MBC transistor may also be referred to as a nanowire transistor or a nanosheet transistor. The introduction of multi-gate devices further increases packing densities of source/drain contacts, gate contacts, and interconnect features connecting thereto. Such increased packing densities call for smaller interconnect features and smaller interconnect features lead to increased current density. Because increased current density increases the rate of

electromigration, it becomes ever more important to implement processes and device structures to prevent electromigration.

[0016] The present disclosure provides interconnect structures and methods to reduce or substantially eliminate electromigration between metal lines and contact vias. Processes according to the present disclosure selectively deposits a metal feature on the contact via such that the contact via is spaced apart from the metal line. Due to the formation of the metal feature, metal migration between the metal line and contact via may be advantageously reduced, and VBDs, TDDb and RC delay due to electromigration may also be reduced. In addition, the contact resistance of the interconnect structure may also be reduced due to the formation of the metal feature.

[0017] The various aspects of the present disclosure will now be described in more detail with reference to the figures. FIG. 1 illustrates a flowchart illustrating a method **100** for forming a semiconductor structure or a portion thereof from a workpiece, according to one or more aspects of the present disclosure. Method **100** is described below in conjunction with FIG. 2, which illustrates a three-dimensional (3D) perspective view of a workpiece **200** undergoing various stages of operations in the method of FIG. 1, and FIGS. 3-14, which are diagrammatic fragmentary cross-sectional views of the workpiece **200** taken along line A-A' at different stages of fabrication according to embodiments of the present disclosure. Method **100** is merely an example and is not intended to limit the present disclosure to what is explicitly illustrated in method **100**. Additional steps can be provided before, during and after the method **100**, and some steps described can be replaced, eliminated, or moved around for additional embodiments of the methods. Not all steps are described herein in detail for reasons of simplicity. Besides what are explicitly shown in figures of the present disclosure, the semiconductor device **200** may include additional transistors, bipolar junction transistors, resistors, capacitors, diodes, fuses, etc. Throughout the present disclosure, like reference numerals denote like features unless otherwise excepted or described. For avoidance of doubts, throughout the figures, the X direction is perpendicular to the Y direction and the Z direction is perpendicular to both the X direction and the Y direction.

[0018] Referring to FIGS. 1-3, method **100** includes a block **12** where a workpiece **200** is provided. Because a semiconductor structure will be formed from the workpiece **200**, workpiece **200** may be referred to as semiconductor device **200** or semiconductor structure **200** in suitable context. Referring to the example of FIGS. 2-3, the semiconductor device **200** includes a substrate **202** and various features formed thereon. In the depicted embodiment, the substrate **202** includes silicon. Alternatively or additionally, the substrate **202** may include another elementary semiconductor, such as germanium (Ge); a compound semiconductor, such as silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb); an alloy semiconductor, such as silicon germanium (SiGe), gallium arsenic phosphide (GaAsP), aluminum indium arsenide (AlInAs), aluminum gallium arsenide (AlGaAs), gallium indium arsenide (GaInAs), gallium indium phosphide (GaInP), and/or gallium indium arsenic phosphide (GaInAsP); or combinations thereof. In some implementations, the substrate **202** includes one or more group III-V materials, one or more group II-IV materials, or combinations thereof. In some implementations, the substrate **202** is a semiconductor-on-insulator substrate, such as a silicon-on-insulator (SOI) substrate, a silicon germanium-on-insulator (SGOI) substrate, or a germanium-on-insulator (GeOI) substrate. Semiconductor-on-insulator substrates can be fabricated using separation by implantation of oxygen (SIMOX), wafer bonding, and/or other suitable methods. The substrate **202** can include various doped regions (not shown) configured according to design requirements of semiconductor device **200**, such as p-type doped regions, n-type doped regions, or combinations thereof. P-type doped regions (for example, p-type wells) include p-type dopants, such as boron (B), gallium (Ga), other p-type dopant, or combinations thereof. N-type doped regions (for example, n-type wells) include n-type dopants, such as phosphorus (P), arsenic (As), other n-type dopant, or combinations thereof. In some implementations, the substrate **202** includes doped regions formed with a combination of p-type

dopants and n-type dopants. An ion implantation process, a diffusion process, and/or other suitable doping process can be performed to form the various doped regions.

[0019] As shown in FIG. 2, the workpiece **200** includes a plurality of fins (or fin elements) **210** disposed on the substrate **202**. In some embodiments, the plurality of fins **210** may be formed from patterning a portion of the substrate **202**. In some alternative embodiments, the plurality of fins **210** may be formed from patterning one or more epitaxial layers deposited over the substrate **202**. An isolation feature **204** is formed between the plurality of the fins **210** to separate adjacent fins. In some embodiments, the isolation feature **204** may include silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), a low-k dielectric, combinations thereof, and/or other suitable materials. In some other embodiments not explicitly shown in the figures, the semiconductor device **200** may be an MBC transistor and the active region may include at least one nanostructure of an MBC transistor. The active region may include first semiconductor layers and second semiconductor layer that are first alternatingly and epitaxially grown on the substrate **202** to form a layer stack. The first semiconductor layer and the second semiconductor layer have different compositions (e.g., Si, SiGe). The semiconductor layer stack having first semiconductor nanostructures and second semiconductor nanostructures is then patterned to form fin-shape stacks of nanostructures. The second semiconductor layers in the channel regions of fin-shape stacks are then selectively removed to release the first semiconductor layers into suspended nanostructures to forming a channel region.

[0020] The workpiece **200** includes gate structures **220** disposed over channel regions of the fins **210**. In the embodiments represented in FIG. 2, the gate structures **220** wrap over channel regions of the fins **210**. While not explicitly shown in the figures, each of the gate structures **220** includes a gate dielectric layer and a gate electrode over the gate dielectric. The gate dielectric layer may include an interfacial layer and a high-k dielectric layer. In some instances, the interfacial layer may include silicon oxide. The high-k dielectric layer is formed of dielectric materials having a high dielectric constant, for example, greater than a dielectric constant of silicon oxide ($k \approx 3.9$). Exemplary high-k dielectric materials for the high-k dielectric layer include hafnium oxide (HfO), titanium oxide (TiO₂), hafnium zirconium oxide (HfZrO), tantalum oxide (Ta₂O₅), hafnium silicon oxide (HfSiO₄), zirconium oxide (ZrO₂), zirconium silicon oxide (ZrSiO₂), lanthanum oxide (La₂O₃), aluminum oxide (Al₂O₃), zirconium oxide (ZrO), yttrium oxide (Y₂O₃), SrTiO₃ (STO), BaTiO₃ (BTO), BaZrO, hafnium lanthanum oxide (HfLaO), lanthanum silicon oxide (LaSiO), aluminum silicon oxide (AlSiO), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), (Ba,Sr)TiO₃ (BST), silicon nitride (SiN), silicon oxynitride (SiON), combinations thereof, or other suitable material. In one embodiment, the high-k dielectric layer is formed of hafnium oxide (HfO). The gate electrode may include multiple layers, such as work function layers, glue/barrier layers, and/or metal fill (or bulk) layers. A work function layer includes a conductive material tuned to have a desired work function (such as an n-type work function or a p-type work function), such as n-type work function materials and/or p-type work function materials. P-type work function materials include TiN, TaN, Ru, Mo, Al, WN, ZrSi₂, MoSi₂, TaSi₂, NiSi₂, WN, other p-type work function material, or combinations thereof. N-type work function materials include Ti, Al, Ag, Mn, Zr, TiAl, TiAlC, TaC, TaCN, TaSiN, TaAl, TaAlC, TiAlN, other n-type work function material, or combinations thereof. A glue/barrier layer can include a material that promotes adhesion between adjacent layers, such as the work function layer and the metal fill layer, and/or a material that blocks and/or reduces diffusion between gate layers, such as the work function layer and the metal fill layer. For example, the glue/barrier layer includes metal (for example, W, Al, Ta, Ti, Ni, Cu, Co, other suitable metal, or combinations thereof), metal oxides, metal nitrides (for example, TiN), or combinations thereof. A metal fill layer can include a suitable conductive material, such as aluminum (Al), copper (Cu), tungsten (W), ruthenium (Ru), titanium (Ti), a suitable metal, or a combination thereof.

[0021] Sidewalls of the gate structures **220** are lined with gate spacers **222**. In some embodiments, the gate spacer **222** may include silicon carbonitride, silicon oxycarbide, silicon oxycarbonitride, or silicon nitride. In some embodiments, a gate replacement or a gate last process may be used to form the gate structures **220**. In an example gate last process, dummy gate stacks are formed over channel regions of the fins **210**. The gate spacers **222** are then deposited over the workpiece **200**, including over sidewalls of the dummy gate stacks. An anisotropic etch process is then performed to recess source/drain regions to form source/drain trenches, leaving behind the gate spacers **222** extending along sidewalls of the dummy gate stacks. After formation of the source/drain trenches, source/drain features (such as the source/drain features **224** in FIG. 3) are deposited into the source/drain trenches in the source/drain regions. The source/drain features **224** may be formed vapor-phase epitaxy (VPE), ultra-high vacuum CVD (UHV-CVD), LPCVD, and/or PECVD), molecular beam epitaxy (MBE), or other suitable epitaxy processes, or combinations thereof. The source/drain features **224** may also be referred to as epitaxial features. Depending on the design of the semiconductor device **200**, source/drain features **224** may be n-type or p-type. When the source/drain features **224** are n-type, they may include silicon (Si) doped with an n-type dopant, such as phosphorus (P) or arsenic (As). When the source/drain features **224** are p-type, they may include silicon germanium (SiGe) doped with a p-type dopant, such as boron (B) or gallium (Ga). In some implementations, annealing processes may be performed to activate dopants in source/drain features **224** of the semiconductor device **200**. A bottom interlayer dielectric (ILD) layer **226** (shown in FIG. 3) is formed over the workpiece **200**. The bottom ILD layer **226** may include, SiO₂, SiN, SiON, tetraethylorthosilicate (TEOS) formed oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), low-k dielectric material ($k < 3.9$), other suitable dielectric material, or combinations thereof. The bottom ILD layer **226** may be deposited using atomic layer deposition (ALD), plasma-enhanced ALD (PEALD), plasma-enhanced chemical vapor deposition (PECVD), and/or other suitable deposition processes. The workpiece **200** is then planarized using a chemical mechanical polishing (CMP) process to expose the dummy gate stacks. The dummy gate stacks are then removed and replaced with the gate structures **220**, the composition of which has been described above.

[0022] With reference to FIG. 3, the workpiece **200** includes a first self-aligned contact dielectric (SAC) layer **234** formed over the gate structure **220** and the gate spacer **222**, a source/drain contact **240** formed over the source/drain feature **224**, and a second self-aligned contact dielectric (SAC) layer **242** formed over the source/drain contact **240**. The formation of the first SAC layer **234** may include multiple processes. For example, the gate structure **220** and the gate spacers **222** are selectively and partially etched to form a recess. In some embodiments, the etching process is selective such that it etches the gate structure **220** and the gate spacers **222** without substantially etching the bottom ILD layer **226**. The etching process may include a dry etch, a wet etch, a combination thereof, or other suitable etching process. In some implementations, the etching process may include at least one component that etches the gate structure at a greater rate than it etches the gate spacer such that a top surface of the gate structure **220** is lower than a top surface of the gate spacer **222** as shown in FIG. 3. The first SAC layer **234** may be then formed in the recess. A thickness of the first SAC layer **234** may be between about 1 nm and about 30 nm. In an embodiment, the first SAC layer **234** is silicon nitride. Other exemplary compositions of the first SAC layer **234** may be formed of silicon oxide, hafnium silicide, silicon oxycarbide, aluminum oxide, zirconium silicide, aluminum oxynitride, zirconium oxide, hafnium oxide, titanium oxide, zirconium aluminum oxide, zinc oxide, tantalum oxide, lanthanum oxide, yttrium oxide, tantalum carbonitride, silicon oxycarbonitride, silicon, zirconium nitride, silicon carbonitride or combinations thereof.

[0023] As depicted in FIG. 3, S/D contacts **240** are formed over the S/D features **224**. The S/D contacts **240** include conductive materials such as Al, W, Cu, or combinations thereof. A silicide layer may be formed between the epitaxial S/D feature **224** and the S/D contact **240**. Formation of

the S/D contacts **240** may involve one or more processes such as lithography, etching, deposition, annealing, and/or planarization processes. In some implementations, source/drain contacts **240** are MEOL features that interconnect FEOL features (for example, epitaxial source/drain features **224**) to BEOL features (for example, the to-be-formed contact via **258** shown in FIG. **6**, the to-be-formed metal line **276** shown in FIG. **11**), thereby electrically and/or physically coupling FEOL features to BEOL features. The second SAC layer **242** is formed on the S/D contact **240**. The material and the processes of forming the second SAC layer **242** may be in a way similar to those used to form the first SAC layer **234**. A thickness of the second SAC layer **242** may be between about 1 nm and about 30 nm.

[0024] Referring to FIG. **1** and FIGS. **4-6**, the method **100** includes a block **14** where a first ILD layer is formed over the workpiece **200** and a block **16** where a S/D contact via is formed over the S/D feature **224**. With reference to FIG. **4**, a capping layer **252** and a first ILD layer **254** are formed over the workpiece **200**. Because the capping layer **252** is disposed over top surfaces of the gate structures **220**, the capping layer **252** may also be referred to as gate-top capping layer **252** or a gate-top etch stop layer **252**. In some embodiments, the capping layer **252** may include silicon carbonitride (SiCN), silicon nitride (SiN), and/or or other suitable materials. In some instances, the first ILD layer **254** includes a thickness along the Z direction and the thickness is between about 11 nm and about 20 nm. The first ILD layer **254** may include an oxide material, such as undoped silica glass (USG), or other suitable material(s). As illustrated in FIG. **5**, one or more openings (such as opening **256**) are formed to expose surfaces of the S/D contacts **240**. In some embodiments, the opening **256** may be formed by penetrating the first ILD layer **254**, the capping layer **252**, and the second SAC layer **242** to expose a portion of the top surface of the S/D contact **240** (e.g., drain contact). Another opening (not shown) may be formed similarly to expose a top surface of another S/D contact **240** (e.g., source contact). In some implementations, one or more rinse or cleaning processes may be performed to clean the exposed S/D contacts **240**, such as the sidewalls through the second SAC layer **242** and the exposed portions of the S/D contacts **240**.

[0025] As depicted in FIG. **6**, one or more S/D contact vias (such as contact via **258**) are formed in the openings (such as opening **256**), respectively, to fill the openings (e.g., the opening **256**). Similar processes may be performed to form another contact via that electrically contacts with the source contact for the source feature of the transistor. In this depicted example, to form the contact via **258**, a glue layer **260** is first conformally deposited over the first ILD layer **254** and into the opening **256** using a suitable deposition technique, such as an ALD process, a PVD process or a CVD process, and then a metal fill layer **262** is deposited over the glue layer **260** using a suitable deposition technique, such as an ALD process, a PVD process or a CVD process. The glue layer **260** may be formed of tungsten, ruthenium, cobalt, copper, molybdenum, titanium nitride, or tantalum nitride. In one embodiment, the glue layer **260** includes tantalum nitride. A thickness of the glue layer **260** may be between about 0.5 nm and about 10 nm. In some embodiments, the glue layer may only include a sidewall portion formed over the sidewall surfaces of the opening **256**. In some embodiments, the glue layer may only include a flat portion formed on the bottom surface of the opening **256**. The metal fill layer **262** may be formed of tungsten, ruthenium, cobalt, copper, molybdenum, aluminum, or an alloy thereof. In one embodiment, the metal fill layer **262** for the contact via **258** may include cobalt. In some embodiments, a width $W_{sub.2}$ along the X direction of the bottommost portion of the metal fill layer **262** may be different from (e.g., smaller than) a width $W_{sub.1}$ (along the X direction) of the topmost portion of the metal fill layer **262**. For example, $W_{sub.1}$ may be between about 2 nm and about 20 nm and $W_{sub.2}$ may be between about 1 nm and about 19 nm.

[0026] Referring to FIG. **1** and FIGS. **7-8**, the method **100** includes a block **18** where a patterned etch stop structure is formed over the contact via **258**. As shown in FIG. **7**, an etch stop structure **263** is formed over the substrate. The etch stop structure **263** may include one or more etch stop layers. When the etch stop structure **263** is a single layer, it may be formed of a nitrogen-containing

dielectric material, such as silicon nitride or silicon carbonitride. In some implementations, a total thickness of the etch stop structure **263** may be between about 2 nm and about 60 nm. When the etch stop structure **263** is multi-layer, as shown in FIG. 7, the etch stop structure **263** may include a first etch stop layer (ESL) **264** and a second ESL **266** formed on the first ESL **264**. With respect to an etch process, the second ESL **266** may have an etch rate that is different from that of the first ESL **264**. This difference in etch rate allows detection of an etch end point. The first ESL **264** and the second ESL **266** may be formed of SiO, SiN, SiCN, SiOC, SiOCN, ZrSi, ZrAlO, TiO, Ta.sub.2O.sub.3, Ta.sub.2O.sub.5, ZrO.sub.2, LaO, ZrN, ZnO, HfO.sub.2, LaO, AlO, AlON, Y.sub.2O.sub.3, TaCN. In some embodiments, the first ESL **264** may be a nitrogen-containing layer and the second ESL **266** may be metal oxide layer. In some embodiments, a thickness of the first ESL **264** is between about 1 nm and about 30 nm and a thickness of the second ESL **266** is between about 1 nm and about 30 nm. With reference to FIG. 8, the etch stop structure **263** is patterned to form a metal line opening **267** for forming metal lines. The formation of the patterned etch stop structure **263** may involve multiple processes such as lithography, etching, and/or cleaning. As shown in FIG. 8, the top surface of the contact via **258** and a portion of the top surface of first ILD layer **254** are exposed in the metal line opening **267**.

[0027] Referring to FIG. 1 and FIG. 9, the method **100** includes a block **20** where a metal feature is selectively formed over the S/D contact via **258** and in the metal line opening **267**. As shown in FIG. 9, a metal feature **268** is selectively formed over the contact via **258** and is spaced apart from the source/drain contact **240** by the second SAC layer **242**, and the capping layer **252**, and the first ILD layer **254**. The metal feature **268** may also be referred to as a metal cap **268** or a conductive cap feature **268** and is formed of a metal different from that of the first metal fill layer **262**. For example, the metal feature **268** may include tungsten, ruthenium, cobalt, copper, molybdenum, or other refractory metals. In some embodiments, a selective deposition process may be implemented such that the metal feature **268** is selectively formed on and self-aligned with the contact via. A thickness T of the metal feature **268** may be between about 1 nm and about 30 nm. If the thickness T is greater than 30 nm, a corresponding manufacturing cost (e.g., cost on the materials for forming the metal feature **268** and/or even materials for forming the to-be-formed metal line **276** in the MLI feature) and time consumed on the formation of the metal feature **268** would be increased. If the thickness T is less than 1 nm, the metal feature **268** may be insufficient to reduce or substantially prevent the electromigration between the contact via **258** and the to-be-formed metal line **276** (shown in FIG. 11). The deposition process may include an ALD process, a CVD process, other suitable methods, or combinations thereof. In this illustrated example, by controlling the deposition process, the metal feature **268** is formed to cover not only the contact via **258** but also a small portion of the top surface of the first ILD layer **254** around the contact via **258** to seal the interface between the contact via **258** and the first ILD layer **254**. The coverage of the metal feature **268** on the small portion of the first ILD layer **254** may be a result of an overgrowth of the metal feature **268** and may further improve device reliability and reduce contact resistance. Along the Z direction, the top surface of the contact via **258** may have, for example, a circular shape, an oval shape, a racetrack shape, a square shape, or a rectangular shape. The metal feature **268** may track the shape of the top surface of the contact via **258** and a distance D between the edge of the metal feature **268** and the edge of the contact via **258** may be between about 3% W.sub.1 (W.sub.1 is shown in FIG. 6) and about 25% W.sub.1, for example. If the distance D is greater than 25% W.sub.1, a parasitic resistance associated with the to-be-formed metal line **276** may be substantially increased. If the distance D is less than 3% W.sub.1, the interfacial seam between the contact via **258** and the first ILD layer **254** may not be fully covered and thus disadvantageously reducing the effectiveness of blocking the electro-migration paths therebetween. In some embodiments, a width W.sub.m (along the X direction) of the metal feature **268** may be between about 3 nm and about 25 nm. The width W.sub.m may also be a diameter of the metal feature **268** when the metal feature **268** has a circular shape.

[0028] In exemplary embodiments, the deposition process of forming the metal feature **268** may include an ALD process that sequentially exposes the top surface of the S/D contact via **258** to two different gaseous precursors in a cyclic manner, i.e., alternating application of a first gaseous precursor and a second gaseous precursor to the top surface of the contact via **258**. The first gaseous precursor may comprise a compound including a conductive element, such as W, Co, Cu, or combinations thereof, and a halogen, such as chlorine (Cl) or fluorine (F). In exemplary embodiments, the first gaseous precursor may be tungsten chloride (WCl₃), tungsten fluoride (WF₆), or other suitable materials. The second gaseous precursor comprises elements such as silicon (Si) and hydrogen (H). Examples of the second gaseous precursor include hydrogen (H₂) and silane (SiH₄). In a specific embodiment, the metal feature **268** includes tungsten, the first gaseous precursor includes WF₆, and the second gaseous precursor includes H₂. The deposition process is implemented at a temperature ranging between about 100° C. and about 550° C. with a processing pressure ranging between about 20 mTorr and 1000 mTorr. In some embodiments, the deposition process may be carried out for about 180 seconds to about 2000 seconds. In some embodiments, a shape of the metal feature **268** may include a convex top surface.

[0029] Referring to FIG. 1 and FIGS. 10-11, the method **100** includes a block **22** where a metal line is formed over the substrate **202** and fills the metal line opening **267**. As illustrated in FIG. 10, a barrier structure **269** is formed over the metal cap **268**. The barrier structure **269** directly contacts both the metal cap **268** and the first ILD layer **254**. In other words, the barrier structure **269** not only covers the top surface of the metal cap **268** but also cover a portion of the top surface of the first ILD layer **254**. As discussed above, the shape of the metal cap **268** may include a convex top surface, thus the barrier structure **269** includes a concave bottom surface. The barrier structure **269** may be a single-layer structure or may include multiple layers and may be formed using a suitable deposition technique, such as an ALD process, a PVD process or a CVD process. When the barrier structure **269** is a single-layer structure, a thickness of the barrier structure **269** may be between about 1 nm and about 60 nm. For example, a thickness of the barrier structure may be between about 5 nm and 20 nm. In this depicted example, the barrier structure **269** includes a first barrier layer **270** and a second barrier layer **272**. For example, the first barrier layer **270** includes a material that prevents metal diffusion from the metal fill layer **274** (shown in FIG. 11) into dielectric layer (e.g., the first ILD layer **254** and ESL structure **263**) and promotes adhesion between the metal fill layer **274** and the dielectric layer, and the second barrier layer **272** includes a material that promotes adhesion between the first barrier layer **270** and the metal fill layer **274**.

[0030] The first barrier layer **270** is conformally deposited to have a generally uniform thickness over the top surface of the workpiece **200** (e.g., having substantially the same thickness on the surfaces of the metal cap **268** and the patterned ESL structure **263**). A thickness of the first barrier layer **270** may be between about 1 nm and about 30 nm. The first barrier layer **270** may be formed of tungsten, ruthenium, cobalt, copper, molybdenum, titanium nitride, or tantalum nitride, and different from the material of forming the metal cap **268**. In some embodiments, the resistivity of the material of the metal cap **268** is less than the resistivity of the material of the first barrier layer **270**. The second barrier layer **272** is conformally deposited to have a generally uniform thickness over the surface of the first barrier layer **270** (e.g., having substantially the same thickness on top and sidewall surfaces of the first barrier layer **270**). A thickness of the second barrier layer **272** may be between about 1 nm and about 30 nm. The second barrier layer **272** may be formed of tungsten, ruthenium, cobalt, copper, molybdenum, titanium nitride, or tantalum nitride and different from the material of forming the first barrier layer **270**. In some embodiments, the resistivity of the material of the metal cap **268** is less than the resistivity of the material of the second barrier layer **272**. Although the second barrier layer **272** is conformally deposited over the first barrier layer **270** in this depicted example, in other embodiments, the second barrier layer **272** may only include a bottom portion that is formed on the top surface of flat portion of the first barrier layer **270**. In some embodiments, the second barrier layer **272** may only include a sidewall portion that is only

formed on sidewall surfaces of the first barrier layer **270**.

[0031] With reference to FIG. **11**, a metal fill layer **274** is formed over the barrier structure **269**. The formation of the metal fill layer **274** may involve multiple processes such as deposition, planarization processes. For example, the metal fill layer **274** is deposited over the barrier structure **269** using a suitable deposition technique, such as an ALD process, a PVD process or a CVD process. The metal fill layer **274** may be formed of tungsten, ruthenium, cobalt, copper, molybdenum, or an alloy thereof. The material of forming the metal fill layer **274** is different from the material of forming the metal cap **268**. In one embodiment, the metal fill layer **274** is formed of ruthenium. A thickness of the metal fill layer **274** may be between about 1 nm and about 50 nm. The combination of the barrier structure **269** and the metal fill layer **274** may be referred as to metal line **276**. In some embodiments, depending on the total thickness of the ESL structure **263**, another ILD layer **278** (as shown in FIG. **12**) may be optionally formed over the ESL structure **263** before the forming of the opening **267** and the metal cap **268**. The composition and processes of forming the ILD layer **278** may be similar to the composition and processes of forming the first ILD layer **254**.

[0032] As discussed above, with the interconnect structures getting smaller, the barrier structure **269** is getting thinner. The thinned barrier structure **269** may not have a sufficient thickness to retard metal migration between the metal line **276** and the contact via **258**. The metal cap **268**, which spaces apart the metal line **276** from the contact via **258**, would advantageously reduce and even substantially eliminate the interdiffusion and metal migration between the metal line **276** and the contact via **258**. In other words, the electron transportation path between the metal line **276** and the contact via **258** is substantially blocked by the metal cap **268**. Moreover, as the overgrowth of the metal cap **268** covers the interfacial seam between the contact via **258** and the first ILD layer **254**, the metal cap **268** may reduce possible electro-migration paths. In some embodiments, the contact via **258** is formed of a first material and the metal line **276** is formed of a second material different from the first material. To reduce the interdiffusion and metal migration between the metal line **276** and the contact via **258**, the metal cap **268**, selectively formed to space apart the metal line **276** from the contact via **258**, is formed of a third material different from the first material and the second material. In some embodiments, the material of the metal cap **268** may be selected such that a solubility between the metal cap **268** and the metal line **276** and a solubility between the metal cap **268** and the contact via **258** are less than a solubility between the metal line **276** and the contact via **258**. As such, the interdiffusion and metal migration between the metal line **276** and the contact via **258** may be further reduced. In some embodiments, the material of the metal cap **268** may be further selected such that the resistivity of the third material is less than the resistivity of that of the first barrier layer **270** and/or the second barrier layer **272**. In one embodiment, the metal fill layer **274** is formed of ruthenium, the metal fill layer **262** is formed of cobalt, and the metal cap **268** is formed of tungsten. Since electromigration between the contact via **258** and the metal line **276** would be advantageously reduced or substantially eliminated, VBDs, TDDb and RC delay due to electromigration may also be advantageously reduced. In addition, because the convex top surface of the metal cap **268** has a greater area than the top surface of the contact via **258**, implementation of the metal cap **268** of the present disclosure may reduce the contact resistance between the metal line **276** and the top surface of the contact via **258**. As such, the device performance is advantageously enhanced.

[0033] Referring to FIG. **1**, method **100** includes a block **24** where further processes may be performed. Such further processes may include forming further structures for interconnecting devices (e.g., the semiconductor device **200**) fabricated across the semiconductor substrate **202**. For example, such further processes may include deposition of an intermetal dielectric (IMD) layer, formation of metal lines, formation of power rails, and/or other suitable semiconductor device features. As discussed above, the semiconductor device **200** may include one or more transistors having one or more gate structures **220**. Such further processes may include forming an

interconnect structure over the gate structure **220**.

[0034] Metal caps similar to the metal cap **268** may also be applied to gate contact vias to reduce or eliminate electro-migration between the contact vias and the metal lines. Reference is made to FIG. **13**. A gate contact via **358** penetrates the first ILD layer **254**, the capping layer **252**, and the first SAC layer **234** and electrically connected to the gate structure **220**. The gate contact via **358** includes a glue layer **360** and a metal fill layer **362**. The glue layer **360** may be formed of tungsten, ruthenium, cobalt, copper, molybdenum, titanium nitride, or tantalum nitride. A thickness of the glue layer **360** may be between about 0.5 nm and about 10 nm. The metal fill layer **362** may be formed of tungsten, ruthenium, cobalt, copper, molybdenum, titanium nitride, or tantalum nitride. A width of the metal fill layer **362** may be between about 1 nm and about 20 nm. In some embodiments, a width $W1'$ of the topmost portion of the metal fill layer **362** may be different from a width $W2'$ of the bottommost portion of the metal fill layer **362**. For example, $W1'$ may be between about 2 nm and about 20 nm and is greater than $W2'$, $W2'$ may be between about 1 nm and about 19 nm. In this depicted example, the gate contact via **358** is formed after the formation of the S/D contact vias **258**, and the gate contact via **358** may be formed of material(s) different from the S/D contact vias (e.g., the contact via **258**). In other implementations, the gate contact via **358** and the S/D contact vias (e.g., the contact via **258**) may be formed simultaneously and formed of same material(s).

[0035] The first ESL **264** and the second ESL **266** are deposited over the workpiece **200**, including over the metal cap **368**. The deposited first ESL **264** and the second ESL **266** are then patterned as described above to form the patterned ESL structure **263**. The patterned ESL structure **263** includes another metal line opening (shown as filled by a metal line **376** in FIG. **13**) and a top surface of the gate contact via **358** is exposed in the another metal line opening. Then, a metal cap **368** may be formed in a way similar to that used to form the metal cap **268** in FIG. **9**. A thickness T' of the metal cap **368** may be between about 1 nm and about 30 nm. In some embodiments, a shape of a top surface of the metal feature **368** may include a convex shape.

[0036] After forming the metal cap **368**, a metal line **376** (e.g., including barrier layers **370**, **372** and a metal fill layer **374**) is deposited into the another metal line opening. In some embodiments, the metal line **376** may be formed in a way similar to that used to form the metal line **276** in FIGS. **10-11**. By providing the metal cap **368** which spaces apart the contact via **358** from the metal line **376**, leakage path between the contact via **358** and the metal line **376** may be substantially blocked and thus electromigration between the metal line **376** and the contact via **358** may be advantageously reduced or substantially eliminated. Therefore, VBDs, TDDb and RC delay due to electromigration may also be advantageously reduced. In addition, the contact resistance at the interface between the metal line **376** and the metal cap **368** is also reduced. As such, the device performance is advantageously enhanced.

[0037] In the above example as discussed with reference to FIGS. **3-13**, both the source contact via and the drain contact via of a transistor in the semiconductor device **200** are frontside contact vias. In some other embodiments, at least one of the S/D contact vias of a transistor may be a backside contact via to accommodate integrated circuits of high functional density. For example, the drain contact via may be a frontside contact via and a source contact via may be a backside contact via. The foregoing metal cap formation processes may also be used to reduce interdiffusion and metal migration between a backside source contact via and a backside power rail. As shown in FIG. **14**, after the frontside interconnect structure is formed at the frontside of the workpiece **200**, a backside interconnect structure would be formed at the backside of the workpiece **200**. The backside processes may be performed by turning the workpiece **200** upside-down.

[0038] The formation of the backside interconnect structure may include multiple processes. For example, at least a portion of the substrate **202** is removed from the backside of the workpiece **200**. Removing at least a portion of the substrate **202** may include various steps such as a planarization process (for example, CMP) and/or an etching process (for example, wet etching, dry etching, or

combinations thereof). In some embodiments, the substrate **202** may be planarized until adjacent semiconductor fin structures are no longer jointed by the substrate **202** or a portion thereof. After the backside of the workpiece **200** is polished, the bottom surface of a source feature **400** (one of the source/drain features **224**) is exposed from the backside. A backside dielectric layer **406** is then deposited covering the bottom surface of the source feature **400**. The backside dielectric layer **406** includes a material similar as that of the bottom ILD layer **226** or the first ILD layer **254**. In some embodiments, the backside dielectric layer **406** may be formed by a deposition process (such as CVD, PVD, ALD), plating, other suitable methods, or combinations thereof. In some embodiments, before depositing the backside dielectric layer **406**, additional processing of the source feature **400** such as forming a silicide region **405** at the bottom portion of the source feature **400** may be performed.

[0039] A backside contact via **407** is then formed in the backside dielectric layer **406** and directly contacts the bottom surface of the source feature **400** (or the silicide region **405**). In this depicted example, the backside contact via **407** includes a glue layer **408** and a metal fill layer **410**. The backside contact via **407** may be self-aligned to the source feature **400** and processes and materials of forming backside contact via **407** may be in a way similar to the formation of the frontside contact via **258** discussed with reference to FIG. 6. A width W_3 (along the X direction) of the backside contact via **407** may be between about 10 nm and about 25 nm and greater than $W_{sub.1}$ ($W_{sub.1}$ is shown in FIG. 6). As such, in some embodiments, the semiconductor device **200** (e.g., a transistor) may include two contact vias (e.g., contact via **258** and contact via **407**) with different dimensions for the corresponding source feature **400** and the drain feature **224**.

[0040] Then, a backside metal cap **412** is formed and electrically contacts the backside contact via **407**. More specifically, the backside metal cap **412** is formed below the backside dielectric layer **406** and directly contacts the bottom surface of the backside contact via **407**. The backside metal cap **412** is formed from a metal different from the metal that forms the metal fill layer **410**. For example, the backside metal cap **412** may include tungsten, ruthenium, cobalt, copper molybdenum, or other refractory metals. The processes and materials of forming backside metal cap **412** may be in a way similar to the formation of the metal cap **268/368** discussed with reference to FIG. 9 and FIG. 13. For example, a selective deposition process may be implemented such that the backside metal feature **412** is substantially selectively formed below and self-aligned with the contact via **407** and is substantially absent from the surfaces of the dielectric layer **406**. In some embodiments, by controlling the deposition process, the backside metal feature **412** is formed to not only cover the contact via **407** but also cover a small portion of a bottom surface of the dielectric layer **406** as discussed with reference to FIG. 9. For example, the backside metal feature **412** may track the shape of the contact via **407** and has a width $W_{m'}$ ranged between about 10 nm and about 35 nm. Since the feature density at the backside of the workpiece **200** may be less than that of the frontside, the backside source contact via **407** may be fabricated to have a width W_3 (along X direction) greater than W_1 to provide better electricity connection. As the backside metal feature **412** tracks the shape of the backside source contact via **407**, the width $W_{m'}$ of the backside metal feature **412** would be greater than the width W_m of the frontside drain contact via **258**. As such, in some embodiments, the semiconductor device **200** may include metal caps (e.g., the frontside drain metal cap **268** and the backside source metal cap **412**) with two different dimensions.

[0041] After the formation of the backside metal cap **412**, a backside power rail **413** is formed below the backside metal feature **412** and electrically connected to the S/D feature **224**. The materials and the fabrication processes of the backside power rail **413** may be similar as the materials and the fabrication processes of the metal line **276**. For example, the backside power rail **413** also includes barrier layers **414**, **416** and a metal fill layer **418**. Due to the formation of the backside metal cap **412**, the backside power rail **413** is spaced apart from the backside source contact via **407** and thus electromigration between the backside power rail **413** and the backside

source contact via **407** may be advantageously reduced or substantially eliminated. In addition, the contact resistance at the interface between the backside power rail **413** and backside source contact via **407** can be advantageously reduced. As such, the device performance is advantageously enhanced. Further fabrication processes may be also performed. For example, backside interconnect structure may be formed to electrically connected to the backside power rail **413**, passivation layers may be formed on the backside of the semiconductor device **200**.

[0042] Other embodiments of methods may be used to form the backside source contact via **407**. For example, after flipping the workpiece **200** up-side-down (not shown), the back side of the workpiece **200** is planarized until the isolation feature **204** (not explicitly shown) is exposed. A patterned hard mask may be used to selectively expose a substrate region that is under the source feature **400** while covering a substrate region that is under the drain feature **224**. The substrate **202** is then anisotropically etched until the bottom surface of the source feature **400** is exposed in a backside source contact opening. The source silicide feature **405** may be formed in the backside source contact opening and directly contacts the source feature **400**. Then, the backside source contact via **407** (including the glue layer **408** and the metal fill layer **410**) would be formed in the opening. For example, various deposition processes may be used to deposit the glue layer **408** and the metal fill layer **410**. A planarization process, such as a CMP process, may follow the deposition process to remove excess materials and provide a planar surface. The materials for forming the backside source contact via **407** may be in a way similar to those of forming the frontside contact via **258**. After the formation of the backside source contact via **407**, in some embodiments, the substrate **202** may be replaced with the backside dielectric layer **406**. The backside metal cap **412** and the backside power rail **413** may be then formed and electrically connected to the backside source contact via **407**. The processes and materials for forming the backside metal cap **412** and the backside power rail **413** may be in a way similar to those of forming the metal cap **268** and the metal line **276**.

[0043] An exemplary semiconductor structure includes a substrate having a source/drain feature, a dielectric layer over the substrate, a source/drain contact via extending through the dielectric layer and electrically connected to the source/drain feature, wherein the source/drain contact via is formed of a first material, a metal feature over the source/drain contact via, wherein the metal feature is formed of a second material different from the first material, and a metal line over the metal feature and electrically connected to the source/drain contact via. The metal line is formed of a third material different from the second material. The source/drain contact via is spaced apart from the metal line by the metal feature.

[0044] In some embodiments, a top surface of the metal feature may directly contact a portion of a bottom surface of the metal line, and a bottom surface of the metal feature may directly contact a top surface of the source/drain contact via. In some embodiments, a shape of a top surface of the metal feature may include a convex shape. In some embodiments, the source/drain contact via may include a glue layer and a metal fill material over the glue layer. In some embodiments, a bottom surface of the metal feature further may cover a portion of a top surface of the dielectric layer.

[0045] In some embodiments, the metal feature may be a first metal feature, the metal line is a first metal line, the semiconductor structure may also include a gate structure formed over the substrate and adjacent to the source/drain feature, a gate contact via extending through the dielectric layer and electrically connected to the gate structure, and a second metal feature disposed between a second metal line and the gate contact via, the second metal line being electrically connected to the gate contact via. The gate contact via may be spaced apart from the second metal line.

[0046] In some embodiments, the second material may also include ruthenium, molybdenum, copper, cobalt, or tungsten. In some embodiments, the metal line may include a barrier layer over the metal feature, wherein the barrier layer is formed of a fourth material different from the second material, and a metal fill layer over the barrier layer. A portion of a bottom surface of the barrier layer may directly contact the metal feature, and another portion of the bottom surface of the

barrier layer may directly contact the dielectric layer. In some embodiments, a resistivity of the second material may be less than a resistivity of the fourth material.

[0047] In some embodiments, the source/drain feature is a drain feature, the dielectric layer is a first dielectric layer, the source/drain contact via is a frontside drain contact via, and the metal feature is a first metal feature. The semiconductor structure may also include a source feature, a second dielectric layer below the source feature, a backside source contact via extending through the second dielectric layer and electrically connected to the source feature, a second metal feature below the backside source contact via, and a backside power rail below the second metal feature and electrically connected to the frontside drain contact via. The backside source contact via may be spaced apart from the backside power rail.

[0048] Another exemplary semiconductor structure includes a contact via embedded in a dielectric layer, a barrier layer over the contact via and the dielectric layer, a metal fill layer over the barrier layer, and a conductive cap feature disposed between the contact via and the barrier layer, wherein the contact via is spaced apart from the barrier layer. The barrier layer is formed of a first material, and the conductive cap feature is formed of a second material, a resistivity of the second material is less than a resistivity of the first material.

[0049] In some embodiments, the barrier layer may directly contact a top surface of the conductive cap feature and a portion of a top surface of the dielectric layer. In some embodiments, the contact via may include a glue layer and a metal fill material over the glue layer. In some embodiments, the semiconductor structure may include a substrate having a source/drain feature and a source/drain contact layer over the source/drain feature. The contact via may extend through the dielectric layer and be electrically connected to the source/drain contact layer.

[0050] In some embodiments, the semiconductor structure may also include a self-aligned contact dielectric (SAC) feature over the source/drain contact layer, and the contact via may extend through the dielectric layer and the SAC feature and directly contact the source/drain contact layer. In some embodiments, the contact via may include cobalt, the metal fill layer may include ruthenium, and the conductive cap feature may include tungsten.

[0051] An exemplary method includes receiving a workpiece including a dielectric layer and a contact via extending through the dielectric layer, selectively forming a metal feature on a top surface of the contact via, forming a barrier layer over the metal feature and the dielectric layer, wherein the contact via is spaced apart from the barrier layer, and, forming a metal fill layer over the barrier layer. The metal feature is formed of a first material and the barrier layer is formed of a second material different from the first material.

[0052] In some embodiments, the selectively forming of the metal feature may include sequentially exposing the top surface of the contact via to a first gaseous precursor and a second gaseous precursor. The first gaseous precursor may include a compound including a conductive element. In some embodiments, the workpiece may include a substrate having a source/drain feature, a source/drain contact over the source/drain feature, and the dielectric layer over the source/drain contact. The contact via may be electrically connected to the source/drain contact. In some embodiments, the first material may include ruthenium, molybdenum, copper, cobalt, or tungsten, and a shape of the top surface of the metal feature may include a convex shape.

[0053] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a transistor; a conductive via extending through a dielectric layer and electrically connected to the transistor; a metal cap disposed on a first surface of the conductive via and a first portion of a second surface of the dielectric layer; and a metal line disposed around the metal cap, wherein the metal line comprises a barrier layer and a metal fill layer embedded in the barrier layer, wherein the metal cap is embedded in the metal line, wherein the barrier layer extends along a second portion of the second surface of the dielectric layer and is spaced apart from the conductive via.
2. The semiconductor structure of claim 1, wherein the conductive via comprises a via filling layer and a glue layer, wherein the glue layer is disposed between the via filling layer and the dielectric layer.
3. The semiconductor structure of claim 1, wherein the barrier layer further extends along a convex surface of the metal cap.
4. The semiconductor structure of claim 1, wherein the conductive via is electrically connected to a gate structure or a source/drain region of the transistor.
5. The semiconductor structure of claim 1, wherein the conductive via is a first conductive via, the metal cap is a first metal cap, and the metal line is a first metal line; wherein the first conductive via is disposed on a first side of the transistor; and wherein the semiconductor structure further comprises: a second conductive via disposed on a second side of the transistor and electrically connected to the transistor, the second side being different from the first side, a second metal cap disposed on a third surface of the second conductive via, and a second metal line around the second metal cap, wherein the second metal cap is embedded by the second metal line.
6. The semiconductor structure of claim 5, wherein the first conductive via is electrically connected to a gate structure of the transistor, and wherein the second conductive via is electrically connected to a source/drain feature of the transistor.
7. The semiconductor structure of claim 1, wherein the metal fill layer is spaced apart from the metal cap by the barrier layer.
8. The semiconductor structure of claim 1, further comprising an etch stop structure disposed on a third portion of the second surface of the dielectric layer, wherein the metal line is embedded in the etch stop structure.
9. A semiconductor structure, comprising: a conductive feature; a dielectric layer disposed over the conductive feature; a conductive via extending in the dielectric layer and electrically connected to the conductive feature; a capping layer disposed over the conductive via and the dielectric layer; and a metal line disposed over the capping layer and the dielectric layer, wherein the capping layer has a bottommost surface, wherein a portion of the bottommost surface extends along a top surface of the dielectric layer.
10. The semiconductor structure of claim 9, wherein the conductive feature comprises a source/drain feature or a gate structure.
11. The semiconductor structure of claim 9, wherein the bottommost surface of the capping layer is wider than a top surface of the conductive via, wherein the top surface of the conductive via is wider than a bottom surface of the conductive via.
12. The semiconductor structure of claim 9, wherein the conductive via comprises a glue layer and a via filling layer over the glue layer, wherein the glue layer is disposed between the via filling layer and the dielectric layer.
13. The semiconductor structure of claim 9, wherein the metal line is separated from the conductive via by the capping layer.
14. The semiconductor structure of claim 9, wherein the conductive feature is a first conductive feature, the dielectric layer is a first dielectric layer, the conductive via is a first conductive via, the

capping layer is a first capping layer, and the metal line is a first metal line; wherein the semiconductor structure further comprises: a second conductive feature disposed adjacent to the first conductive feature, a second dielectric layer disposed below the second conductive feature, a second conductive via extending in the second dielectric layer and connected to the second conductive feature, a second capping layer disposed below the second conductive via and the second dielectric layer, wherein the second capping layer is connected to the second conductive via, and a second metal line disposed below the second capping layer and the second dielectric layer.

15. The semiconductor structure of claim 14, wherein the first conductive feature comprises a gate structure, and wherein the second conductive feature comprises a source/drain feature.

16. A semiconductor structure, comprising: transistors; a frontside conductive via disposed over and electrically connected to the transistors; a frontside cap layer disposed on a top surface of the frontside conductive via, wherein the frontside cap layer has a bottom surface wider than the top surface of the frontside conductive via; a frontside metal line disposed over the frontside cap layer; a backside conductive via disposed below and electrically connected to the transistors; a backside cap layer disposed below a bottom surface of the backside conductive via, wherein the backside cap layer has a top surface wider than the bottom surface of the backside conductive via; and a backside metal line disposed below the backside cap layer.

17. The semiconductor structure of claim 16, wherein the frontside metal line is disposed over a curved top surface of the frontside cap layer.

18. The semiconductor structure of claim 16, wherein the frontside conductive via is electrically connected to a gate electrode of the transistors, and wherein the backside conductive via is electrically connected to a source/drain feature of the transistors.

19. The semiconductor structure of claim 16, wherein the top surface of the backside cap layer is wider than the bottom surface of the frontside cap layer.

20. The semiconductor structure of claim 16, wherein the bottom surface of the backside conductive via is wider than the top surface of the frontside conductive via.
